Sadaf Siddiqui

Senior AI Engineer | Generative AI Developer | Full-Stack AI Engineer

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Professional Summary

Senior AI Engineer with over 10 years of full-stack development experience, now building production generative AI systems end to end. Currently developing Scout, an AI assistant for a K-12 curriculum platform, using Amazon Bedrock, LangGraph, Pinecone RAG, FastAPI on AWS Lambda, Vue.js/Nuxt and Terraform. Experienced in RAG pipelines, agentic workflows, prompt engineering, LLM guardrails, serverless architecture and response streaming. Certified Scrum Product Owner (CSPO) with a record of leading cross-functional teams and mentoring developers.

Technical Skills

- **Generative AI and LLMs:** Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM Guardrails, Model Optimization, Response Streaming, Amazon Bedrock (Nova Micro), GPT-4o, Gemini, DeepSeek R1, OpenAI API
- **Agentic AI Frameworks:** LangGraph, LangChain, CrewAI, Tool Calling, Multi-Agent Systems
- **AI Coding Tools:** Claude Code, GitHub Copilot
- **Vector Databases:** Pinecone, PGVector, FAISS
- **Languages:** Python, TypeScript, JavaScript, PHP, SQL
- **Backend and APIs:** FastAPI, Node.js, NestJS, Express, GraphQL, REST APIs
- **Frontend:** Vue.js, Nuxt, React, Next.js, Redux, Tailwind CSS
- **Cloud and Serverless:** AWS Lambda, Amazon API Gateway, Lambda Function URLs, AWS Lambda Web Adapter, Amazon DynamoDB, Amazon CloudWatch, Amazon S3, AWS Amplify, GCP
- **Infrastructure as Code and DevOps:** Terraform, CI/CD, Jenkins, Git, GitHub, GitLab
- **Testing:** Unit Testing with Jest and Enzyme, End-to-End Testing with Playwright, Test-Driven Development
- **Databases:** MongoDB, PostgreSQL, MySQL, DynamoDB

- **Practices:** Agile/Scrum, Code Review, Observability, Technical Leadership, Mentoring

Professional Experience

Senior Software Engineer (AI) | Code Connect | Remote | Dec 2024 – Present

Project: Scout, an AI assistant for Studies Weekly Online (SWO), a K-12 educational curriculum platform

- Built Scout end to end: a LangGraph agent on Amazon Bedrock (Nova Micro) with Pinecone RAG over curriculum content, a FastAPI backend on AWS Lambda, and a Vue.js/Nuxt frontend.
- Designed Amazon Bedrock Guardrails for K-12 appropriate content, versioned each guardrail change in Terraform, and kept an incident log built from CloudWatch model invocation logs.
- Engineered prompt routing with lexical trigger-word rules and context trimming to prevent instruction drift, making a small, low-cost model reliable in multi-turn conversations.
- Provisioned infrastructure with Terraform, including a multi-route API Gateway in front of a single Lambda and a Pinecone ingestion CLI for curriculum content.
- Leading migration from buffered API Gateway responses to real-time streaming with Lambda Function URLs, AWS Lambda Web Adapter, Bedrock converse_stream and LangGraph astream_events.
- Wrote the eight-phase migration specification, covering a DynamoDB schema, Terraform modules with decommission plans for legacy Lambdas, and CloudWatch metrics with time to first token as the primary service-level objective.
- Built page-aware assistant features in Nuxt: a client plugin that collects page context from the DOM and a UI action tool that highlights the buttons a user needs.
- Set up local development with DynamoDB Local and a scenario-based Bedrock emulator so the team can test without cloud calls, and wrote end-to-end tests with Playwright.
- Performed prompt engineering and model evaluation across GPT-4o, Gemini, DeepSeek R1 and Amazon Bedrock foundation models, and coordinated sprint planning and the backlog for RAG features.

Lead Developer | Workroom.build | Remote | Feb 2024 – Dec 2024

- Led a cross-functional, multinational team building a generative AI proof of concept, coordinating technical and business stakeholders.
- Engineered a multi-model AI agent for complex task automation using LangChain, FastAPI, Next.js, OpenAI and Google Gemini models (Gemini 1.5 Flash, Gemini 1.5 Pro), and deployed it on Google Cloud Platform (GCP) using Google Compute Engine and Google Cloud Storage buckets.
- Defined product vision and ran product discovery sessions to identify market opportunities for AI-powered web applications.
- Integrated third-party APIs for data scraping, search and visualization.
- Facilitated agile ceremonies, including sprint planning, standups and retrospectives.

Tech Lead | Groupshop | Remote | Jan 2021 – Feb 2024

- Managed the product development lifecycle for a custom Shopify application, delivering within tight deadlines.
- Designed a scalable architecture using Node.js, NestJS, TypeORM, GraphQL and MongoDB. Deployed the backend on AWS Elastic Beanstalk with a CI/CD pipeline and the frontend on AWS Amplify.
- Applied test-driven development in React, reducing production errors by 15%.
- Mentored developers and improved code quality through structured code reviews.

Senior MERN Developer | ENN Technologies | Remote | Jan 2020 – Jan 2021

- Designed and deployed responsive web applications using MongoDB, Express, React and Node.js.
- Developed a reusable React component library with Storybook, making UI upgrades and maintenance faster and more consistent.
- Wrote unit tests with Jest and Enzyme for the SearchSmartly React frontend.
- Optimized database queries for a 10% efficiency gain and led performance work that improved application performance by 10%.
- Deployed Packrit & Searchsmartly on AWS EC2 and AWS Amplify, and led client requirements-gathering sessions.

& Searchsmartly

Senior Development Engineer | Auto360 | Remote | Jul 2019 – Jan 2020

- Led mobile app architecture and cross-functional collaboration, improving performance by 20%.

- Managed the technical roadmap and feature prioritization with stakeholders, including scalability and data security strategy.

Senior Web Developer | InGenious | Remote | Jan 2016 – Jan 2019

- Built scalable web applications with PHP, Laravel and JavaScript, accelerating development timelines by 25% with reusable code libraries.
- Worked with QA and UX teams to improve product usability.

Earlier Experience | 2006 – 2016

- Web Technologies Professional, BusinessJot (Nov 2013 – Dec 2016)
- Senior Software Developer, Kolachi Advanced Technologies (Jun 2012 – Jan 2013)
- Software Engineer, KalSoft (Jul 2011 – Apr 2012)
- Web Developer, Right Solutions (Jul 2010 – Jun 2011)
- Web Developer, Logo Design Pvt Ltd (Jun 2008 – Jul 2010)
- Web Developer, Pakservice Network (Mar 2006 – Jun 2008)

Selected Projects

- **Scout (Studies Weekly Online):** AI assistant for a K-12 curriculum platform with RAG, guardrails, page-aware UI help and streaming responses. Tech: Amazon Bedrock, LangGraph, Pinecone, FastAPI, AWS Lambda, DynamoDB, MongoDB, Vue.js, Nuxt, Terraform, CloudWatch.
- **Workroom.bot:** AI agents that enrich lists, research companies and process hundreds of documents through a table interface. Tech: Next.js, Chakra UI, FastAPI, LangChain, OpenAI, SQLAlchemy, PostgreSQL.
- **Custom Shopify App:** Version 2 of an upsell app integrated with the Shopify GraphQL API. Tech: Next.js, TypeScript, NestJS, TypeORM, MongoDB, GraphQL.
- **Packrit:** E-commerce website with test-driven development and Node.js APIs. Tech: React, Redux Toolkit, Material UI, Node.js, Jest, AWS Amplify, EC2.
- **SearchSmartly:** Real estate platform built with a UK-based team. Tech: React, Redux, Node.js, Express, Sequelize, PostgreSQL.

Education

- **Master of Science in Computer Science** | University of Karachi | 2009
- **Bachelor of Science in Computer Science** | University of Karachi | 2005

Certifications

- Certified Scrum Product Owner (CSPO), Scrum Alliance | Jul 2025 – Jun 2027
- Introduction to LangGraph, LangChain | Oct 2024
- ChatGPT Prompt Engineering for Developers, DeepLearning.AI | Mar 2024
- LangChain Chat with Your Data, DeepLearning.AI | Mar 2024
- LangChain for LLM Application Development, DeepLearning.AI | Mar 2024
- Advanced React, Meta | Jun 2023
- Building Conversational Experiences with Dialogflow, Coursera | May 2020
- Deep Learning with TensorFlow 2.0, 365 Data Science | Apr 2020